



30 June 2026

TRIDENT-GF180-TESTSTRUCTURES Datasheet

1 Known issues

1. The markings of pin 100, pin 1 and pin 2 are incorrect. Pin 100 marking is actually pin 99. Pin 1 marking is pin 100. Pin 2 marking is pin 1.

2 Applications

- Leakage evaluation of GlobalFoundries GF180MCU (`gf180mcuD`) 180 nm process
- Evaluation of guard-trace effectiveness
- Evaluation of foundry vs. custom guarded ESD pad approaches for ultra-low current measurement
- Characterization of discrete HV/LV NFET and PFET leakage under guarding
- Characterization of MIM capacitor leakage and perimeter effects

- Evaluation of picoampere two-transistor charge-pump / current generator structure

3 Description

TRIDENT-GF180-TESTSTRUCTURES is a dedicated test ASIC to assess feasibility of ultra-low current (pA and below) measurements in `gf180mcuD`. It is available open source [here](#). The primary metric is control and predictability of leakage current from “hot” measurement nodes. Primary leakage paths of interest are ESD protection networks, transistor, MIM capacitor, and leakage through insulator mitigated by driven guards.

Pins between structures are often interleaved, to allow for the neighboring pins to be driven as guard. This is especially useful when large voltage between pins are expected on the same device, such as the MIM capacitors.

4 Block Diagram

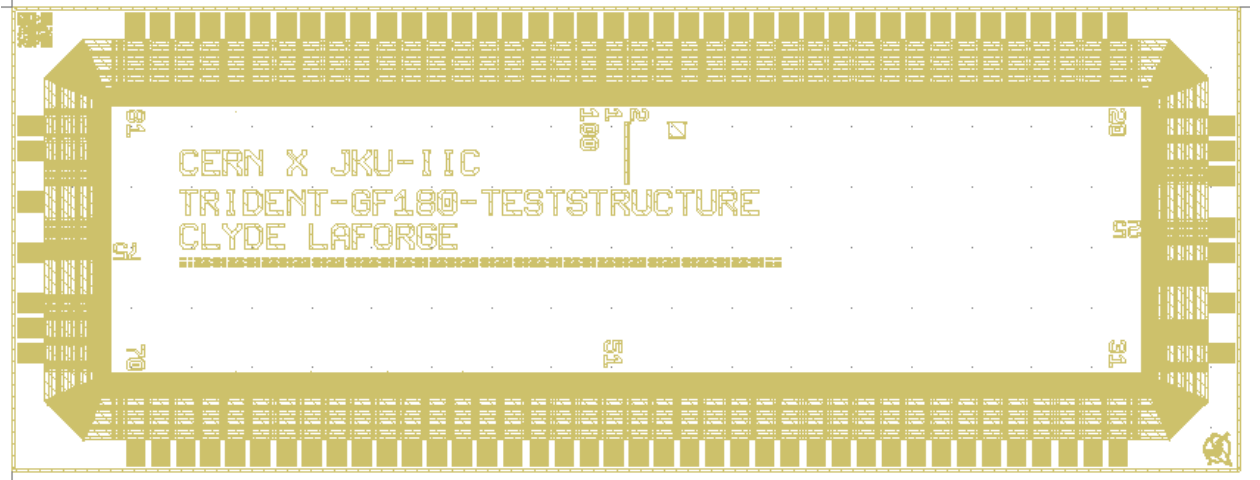


Figure 1: Top level view of the chip

5 Pin Configuration and Function Descriptions

#	Pin name	Decription
1	MC_MAX_P1	MIM capacitor, max perimeter, terminal P1
2	MC_MIN_P2	MIM capacitor, min perimeter, terminal P2
3	MC_MIN_P1	MIM capacitor, min perimeter, terminal P1
4–7	UNUSED	Do not connect
8	MC_GUARD_P2	Guard for MIM capacitor terminal P2
9–13	UNUSED	Do not connect
14	GL_TOP	Guard leakage structure, top guard
15	GL_BOT	Guard leakage structure, bottom guard
16	GL_SIDE	Guard leakage structure, side guard
17	GL_HOT	Guard leakage structure, hot node
18	GL_PAD_GUARD	Guard leakage structure, guard of hot bondpad
19	VSS	Ground supply
20	VDD	Positive supply
21	PESD_GS_HOTN	Pad ESD (guard+sense), inverted hot output
22	PESD_GS_SENSEN	Pad ESD (guard+sense), inverted sense output
23	PESD_GS_SENSE	Pad ESD (guard+sense), sense input
24	NO_BOND	No bondpad present, do not bond
25	PESD_GS_HOT	Pad ESD (guard+sense), hot input
26	NO_BOND	No bondpad present, do not bond
27	PESD_G_HOTN	Pad ESD (guard), inverted hot output
28	NO_BOND	No bondpad present, do not bond
29	PESD_G_HOT	Pad ESD (guard), hot input
30	PESD_GUARD	Guard for all PESD structures
31	VSS	Ground supply
32	VDD	Positive supply

33–48	UNUSED	Do not connect
49	DESD_VDD_K	Discrete ESD diodes to VDD, cathode
50	DESD_VSS_K	Discrete ESD diodes to VSS, cathode
51	DESD_VDD_A	Discrete ESD diodes to VDD, anode
52	DESD_GUARD	Guard for discrete ESD diode structures
53	VSS	Ground supply
54	VDD	Positive supply
55	TR_GUARD	Transistor structures, guard (common)
56	TR_HP_D	High-voltage PFET, drain
57	TR_HP_G	High-voltage PFET, gate
58	TR_HP_S	High-voltage PFET, source
59	TR_HN_D	High-voltage NFET, drain
60	TR_HN_G	High-voltage NFET, gate
61	TR_HN_S	High-voltage NFET, source
62	TR_LP_D	Low-voltage PFET, drain
63	TR_LP_G	Low-voltage PFET, gate
64	TR_LP_S	Low-voltage PFET, source
65	TR_LN_D	Low-voltage NFET, drain
66	TR_LN_G	Low-voltage NFET, gate
67	TR_LN_S	Low-voltage NFET, source
68	TR_GUARD	Transistor structures, guard (common)
69	VSS	Ground supply
70	VDD	Positive supply
71	PL_GUARD	Pad leakage structure, guard (common)
72	PL_FLOAT	Pad leakage structure, floating pad
73	PL_GS_SENSE	Pad leakage (guard+sense), sense input
74	NO_BOND	No bondpad present, do not bond
75	PL_GS_HOT	Pad leakage (guard+sense), hot input
76	NO_BOND	No bondpad present, do not bond
77	PL_G_HOT	Pad leakage (guard), hot input
78	NO_BOND	No bondpad present, do not bond
79	PL_A_HOT	Pad leakage (analog), hot input
80	PL_GUARD	Pad leakage structure, guard (common)
81	VSS	Ground supply
82	VDD	Positive supply
83	CP_GUARD	Charge pump structure, guard
84	CP_G2	Charge pump, gate 2
85	CP_VS	Charge pump, VS node (drain+source)
86	CP_G1	Charge pump, gate 1
87	CP_VW	Charge pump, VW node (bulk)
88–96	UNUSED	Do not connect
97	VSS	Ground supply
98	VDD	Positive supply
99	MC_GUARD.P1	Guard for MIM capacitor terminal P1

100	MC_MAX_P2	MIM capacitor, max perimeter, terminal P2
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UNUSED pads are foundry-provided analog ESD pad with their output left unconnected.

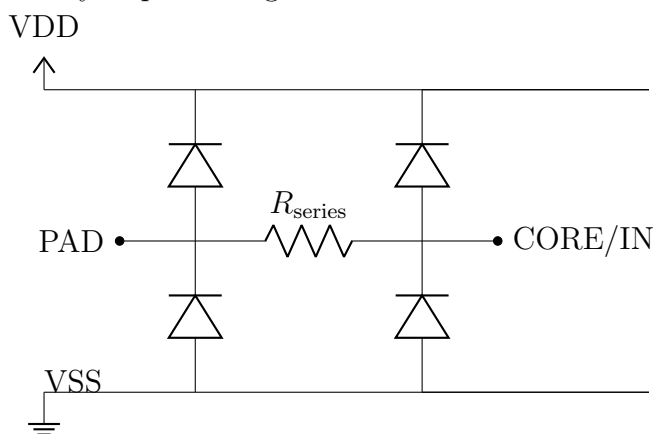
NO_BOND pads have not actual bondpad implemented. They are missing due to the underlying structures being too large. Instead of simply removing them from lists, they are kept nonetheless so that the pad positions and numbering stays regular.

6 ESD protection pads

6.1 Foundry provided analog pad

Below is a diagram of the foundry provided analog input pad, which provide a classic protection structure. The input diodes (on the pad side) are very large as they are supposed to take the brunt of the ESD pulses. The series resistor and output diodes are optional: they are not provided within the pad itself, but users are encouraged to add them in the documentation. The value of the series resistor is typically a few hundred ohms. This allow a secondary path for the current in case of an ESD spike, and the series resistor allows for a reduction of the voltage to which the core will be exposed.

This pad has an implementation with no output, to characterize its leakage in ADDREF. It does not have an implementation with output connected to an inverted to determine its capability of protecting the core transistors due to a lack of time during the layout process.



6.2 Custom guard pad

This adds a guard trace around the input trace. This allows for the internal diodes (between PAD and GUARD) to experience a near zero voltage drop, and hence have minimal leakage. The guard voltage could be driven by a buffer connected to the core voltage. It could also be driven independently from the input (if it is at a fixed voltage), and trimmed by-device for a specific operating point, likely temperature.

In order to provide somewhat comparable ESD protection to the foundry provided analog pad, the size of each diode is doubled compared to it.

As the inner VSS diode is not connected directly to VSS, it is necessary to put it in a deep N-well.

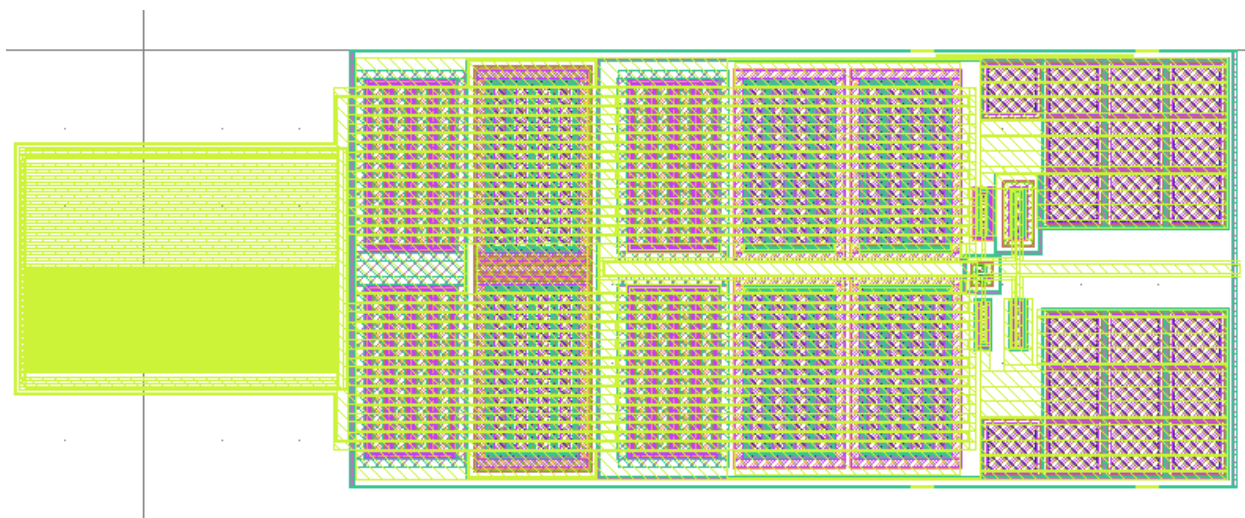
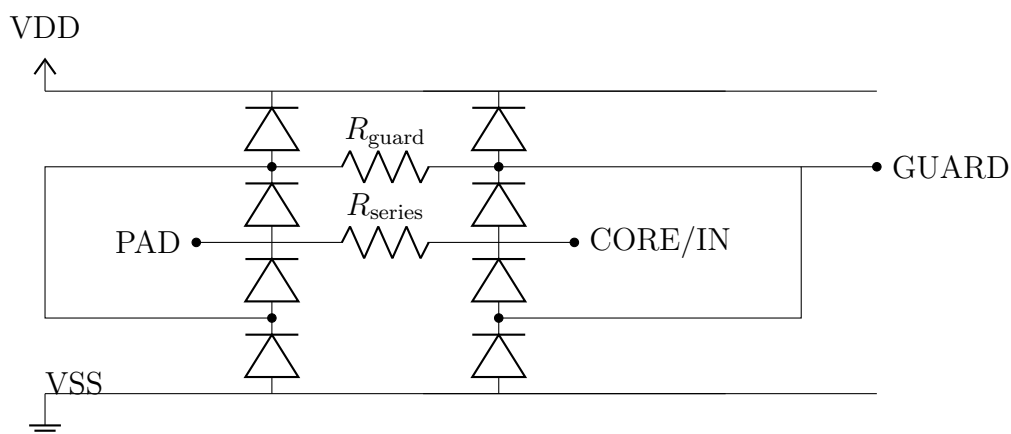


Figure 2: A graphic representation of the layout of the custom guard pad



6.3 Custom guardsense pad

The custom guardsense pad goes one step further than the custom guarded pad. It duplicates the internal components (zero-voltage drop diodes and series resistor), and hence provide 2 new ports, named "SENSE" and "CORE SENSE". Because the implementation of the sense and pad is fully symmetric and interleaved, any leakage experience on the PAD line should be the same on the sense line, provided minimal voltage drop on the series resistor.

This allows for sensing the leakage of the pad input from the core sense. In turn, a the guard voltage can be dynamically adjusted for zero leakage.

For the layout, it was necessary to implement custom ESD diodes, so that it was possible to interleave the guard and hot signals.

6.4 Custom no ESD guarded pad

This is simply a bond pad with a guarded metal output. No device whatsoever are connected to the input, which means that it has no ESD protection, but also minimal leakage on its own.

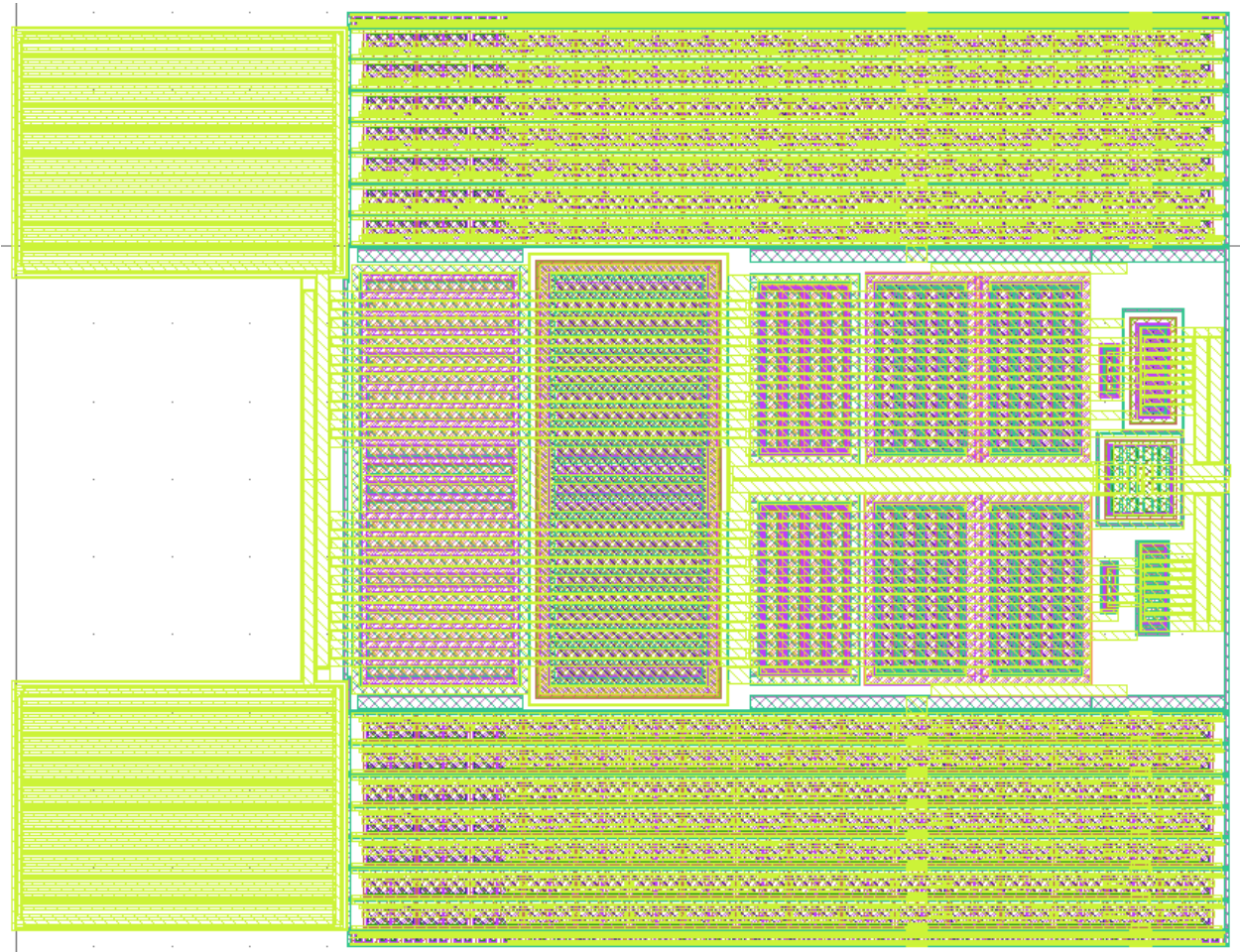


Figure 3: A graphic representation of the layout of the custom guardsense pad

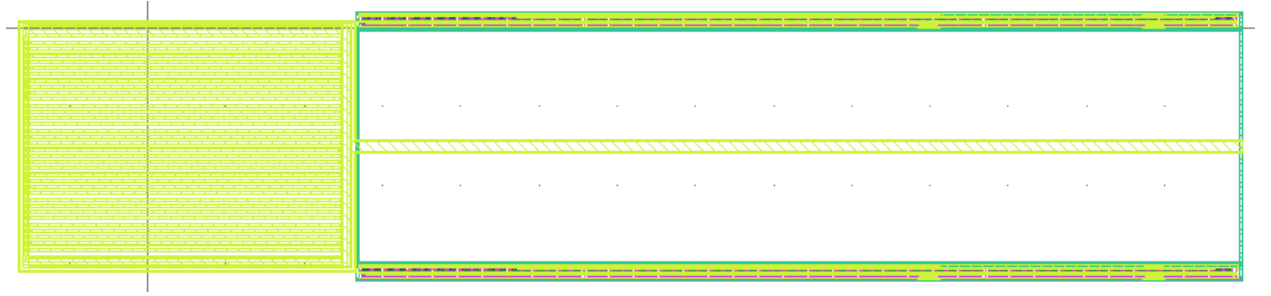


Figure 4: A graphic representation of the layout of the custom no esd pad

This pad is used anytime there is a device characterization pin, so that the measurement correspond to the actual device, and not the ESD protection circuitry. In order to comply with antenna rules, it has sometime been necessary to route the signal up and down to upper metal layers to comply with antenna rules. An unconnected custom no ESD guarded pad is implemented in 7.9, such that it will be possible to have a reference to other measurements.

7 Device Descriptions (by Device)

7.1 MIMCAP_MAX_PERIM

7.1.1 Description

MIM capacitor of ($256 \times 16\mu\text{m}$ and 8192fF). Its high perimeter geometry is to study the effect of perimeter in the MIM capacitor leakage. It will be compared against 7.2.

7.1.2 Pins (array)

#99,	Guard
#100	Pin 2
#1	Pin 1

Pin 1 connects to the lower plate, pin 2 to the upper plate.

7.1.3 Diagram

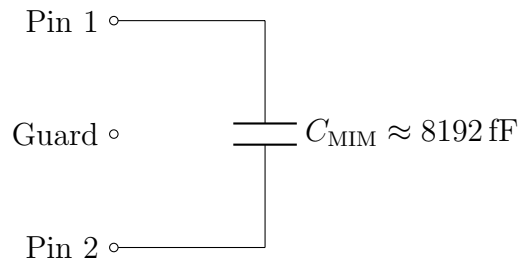


Figure 5: MIM capacitor (MAX_PERIM) with guard.

7.2 MIMCAP_MIN_PERIM

7.2.1 Description

MIM capacitor of ($64 \times 64 \mu\text{m}$ and 8192 fF). Its minimal perimeter geometry is to study the effect of perimeter in the MIM capacitor leakage. It will be compared against 7.1.

7.2.2 Pins (array)

```
#2 Pin 2
#3 Pin 1
#4 Guard
```

Pin 1 connects to the lower plate, pin 2 to the upper plate.

7.2.3 Diagram

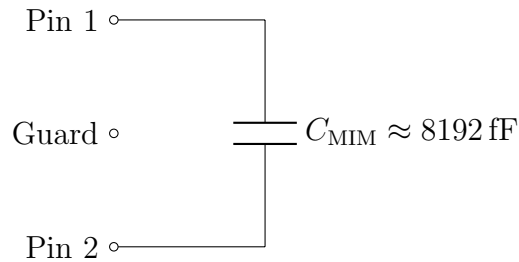


Figure 6: MIM capacitor (MIN_PERIM) with guard.

7.3 GUARD_LEAKAGE

7.3.1 Description

This is a guarded line of $1041 \times 3 \mu\text{m}$ in metal XXX. It is guarded independently on the top, bottom and sides. These guard planes can be swept in relation to the hot so that the influence of guarding in silicon can be measured.

7.3.2 Pins (array)

```
#14 Top guard plane
#15 Bottom guard plane
#16 Side guard traces
#17 Hot
#18 Guard of the hot bondpad
```

7.3.3 Diagram



7.4 PAD_ESD_GUARDSENSE

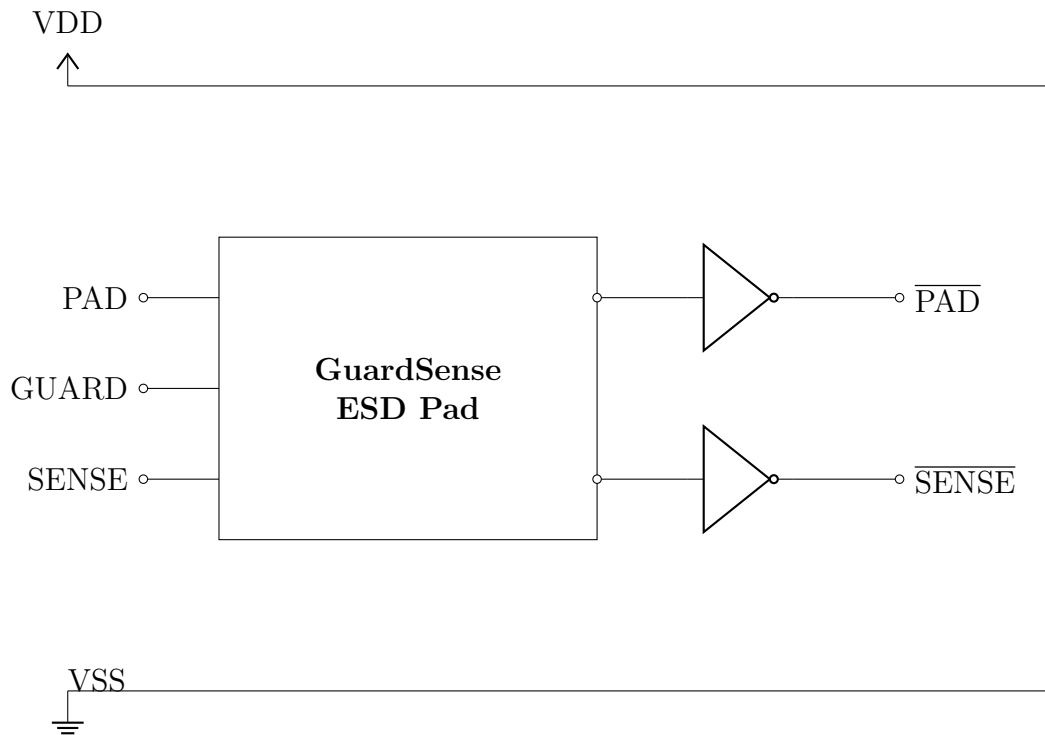
7.4.1 Description

Evaluate guard+sense approach and the ability to trim guard voltage using sense leakage. See 6.3 for details on the internal of the guardsense pad. The purpose of the inverters is to measure up to what sort of ESD pulse the gate of the internal low voltage transistors making up the inverters can survive.

7.4.2 Pins (array)

#21	Inverted hot output
#22	Inverted sense output
#23	Sense input
#25	Hot input
#30	Guard

7.4.3 Diagram



7.5 PAD_ESD_GUARD

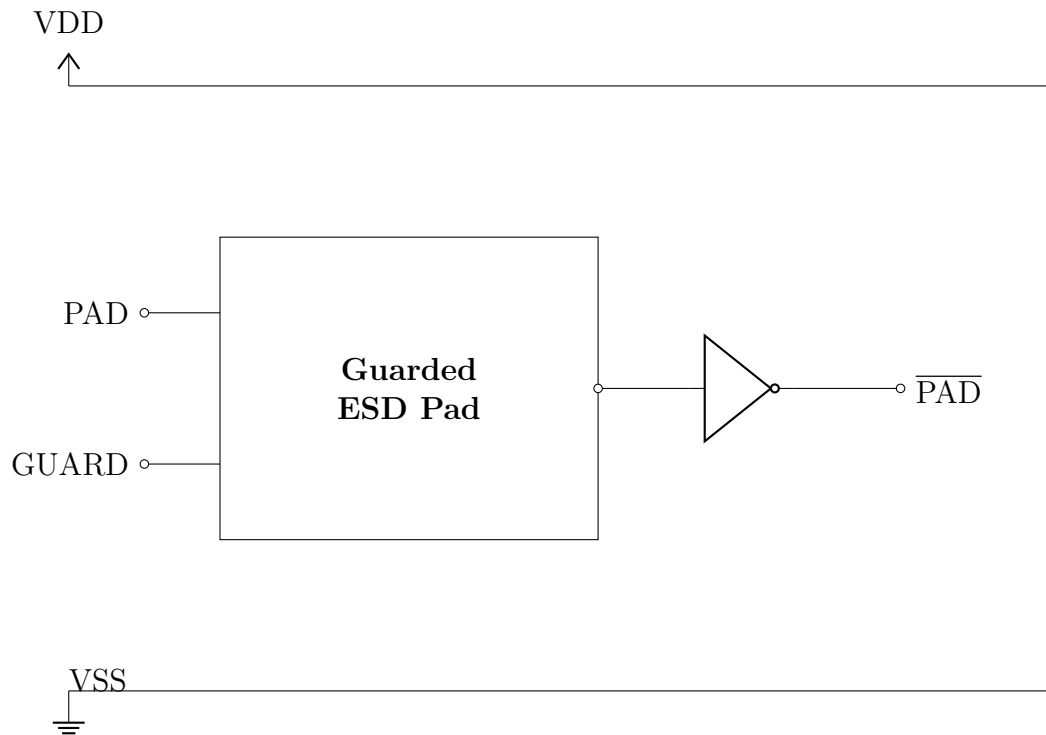
7.5.1 Description

Baseline custom guarded ESD pad without sense line. The purpose of the inverter is to evaluate the robustness of the internal low-voltage inverter gate under ESD stress at the pad input.

7.5.2 Pins (array)

#27 Inverted hot output
 #29 Hot input
 #30 Guard

7.5.3 Diagram



7.6 ESD_DIODE_VDD

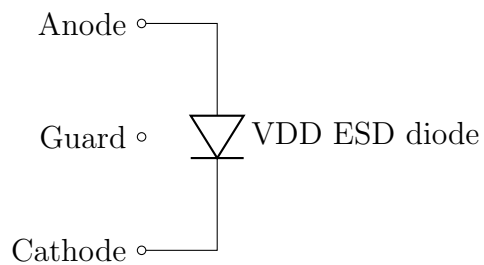
7.6.1 Purpose

Standalone foundry-provided vdd esd diode for ultralow current I-V characterization.

7.6.2 Pins (array)

#49 Cathode
#51 Anode
#52 Guard

7.6.3 Diagram



7.7 ESD_DIODE_VSS

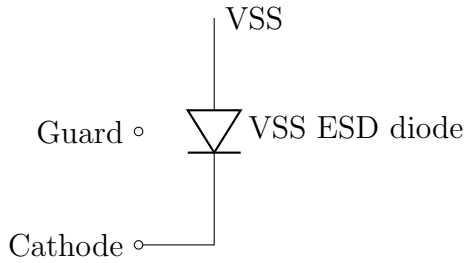
7.7.1 Purpose

Standalone foundry-provided VSS ESD diode for ultralow current I-V characterization. The anode is shorted to VSS through the substrate.

7.7.2 Pins (array)

#50 Cathode
#52 Guard

7.7.3 Diagram



7.8 Guarded HV/LV NFET/PFET, W=0.75um L=0.7um

7.8.1 Description

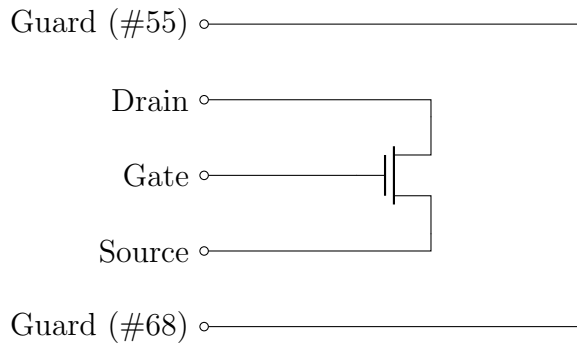
Direct access to D/G/S for transistor leakage characterization under guarding. No ESD pads were used on signal of interest, but an up and down metal break is made to comply with antenna rules.

7.8.2 Pins (array)

Device type	Drain	Gate	Source
HV PFET:	#56	#57	#58
HV NFET:	#59	#60	#61
LV PFET:	#62	#63	#64
LV NFET:	#65	#66	#67

Pin #55 and #68 (internally shorted) as used as guard for all transistor structures.

7.8.3 Diagram (generic, horizontal MOS symbol)



7.9 PAD_LEAK_EMPTY

7.9.1 Description

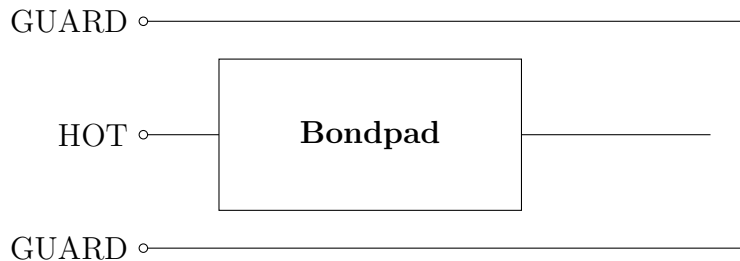
Custom no ESD bondpad, presented in 6.4 connected to nothing. It will be used as a reference to all other measurement involving this pad.

7.9.2 Pins (array)

#71 Guard
#72 Hot
#80 Guard

Guards are internally shorted together.

7.9.3 Diagram



7.10 PAD_LEAK_GUARDSENSE

7.10.1 Purpose

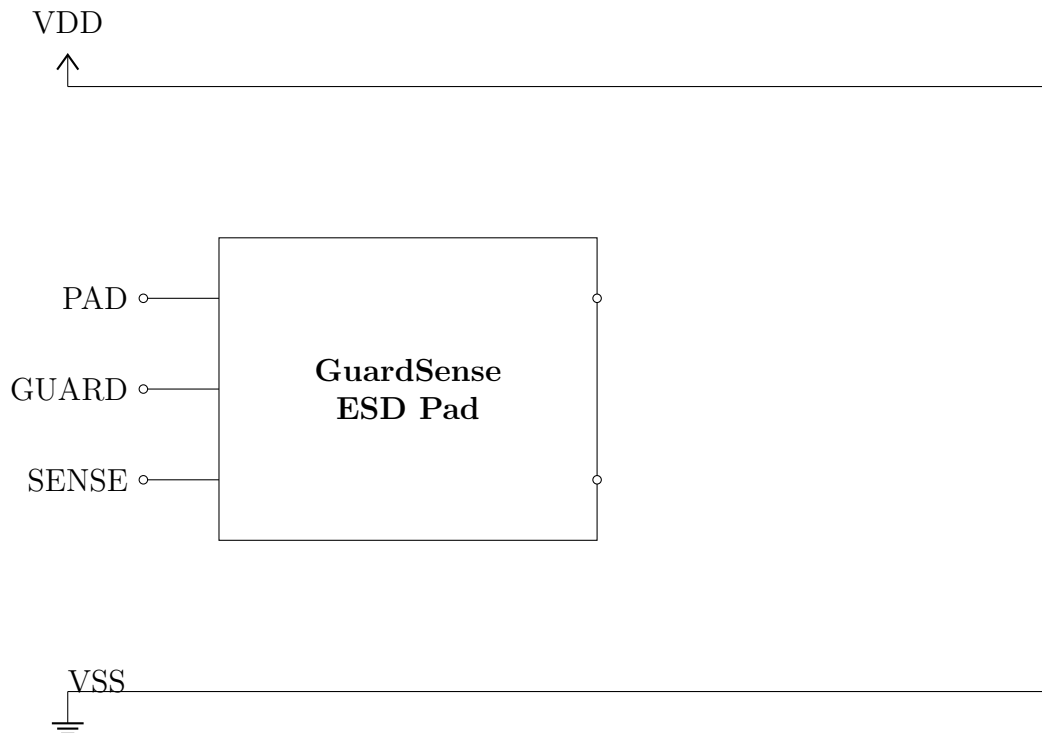
Implements the custom guardsense pad with nothing connected to its output. This allows for measuring the intrinsic leakage of the device. It could also be used for ESD testing, to characterize the robustness of the pad structure itself, as opposed to 7.4, which characterizes its ability to protect the core transistors.

7.10.2 Pins (array)

#71 Guard
 #73 Sense input
 #75 Hot input
 #80 Guard

Guards are internally shorted together.

7.10.3 Diagram



7.11 PAD_LEAK_GUARD

7.11.1 Purpose

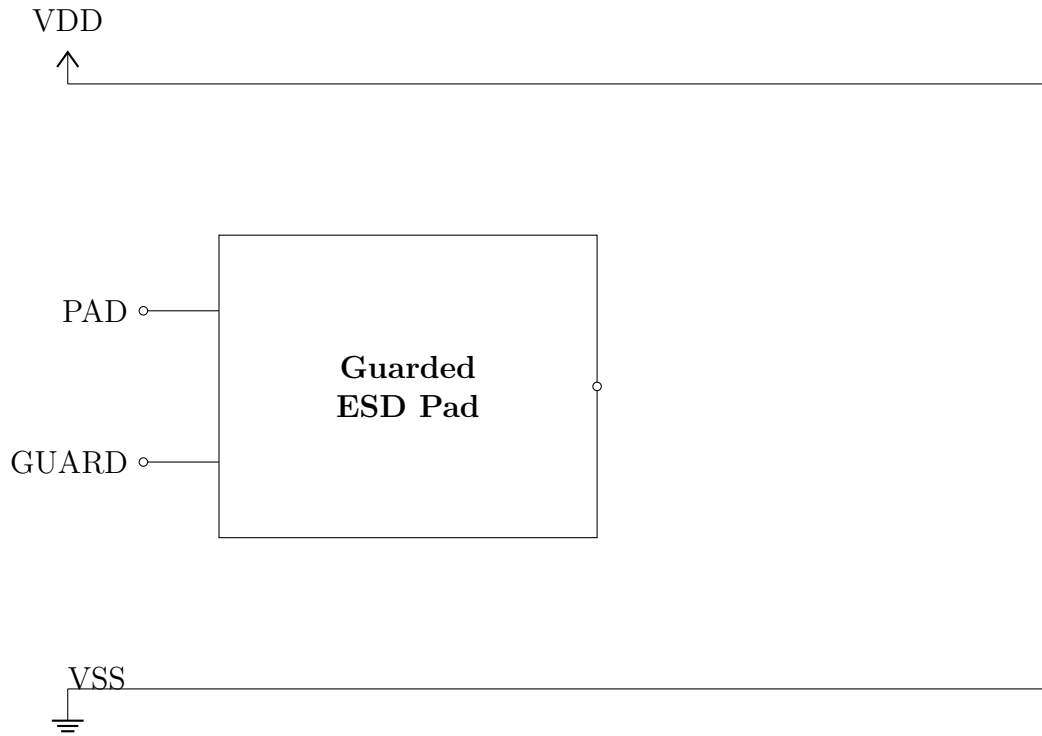
Implements the custom guard pad with nothing connected to its output. This allows for measuring the intrinsic leakage of the device. It could also be used for ESD testing, to characterize the robustness of the pad structure itself, as opposed to 7.5, which characterizes its ability to protect the core transistors.

7.11.2 Pins (array)

#71 Guard
 #77 Hot input
 #80 Guard

Guards are internally shorted together.

7.11.3 Diagram



7.12 PAD_LEAK_ASIG

7.12.1 Description

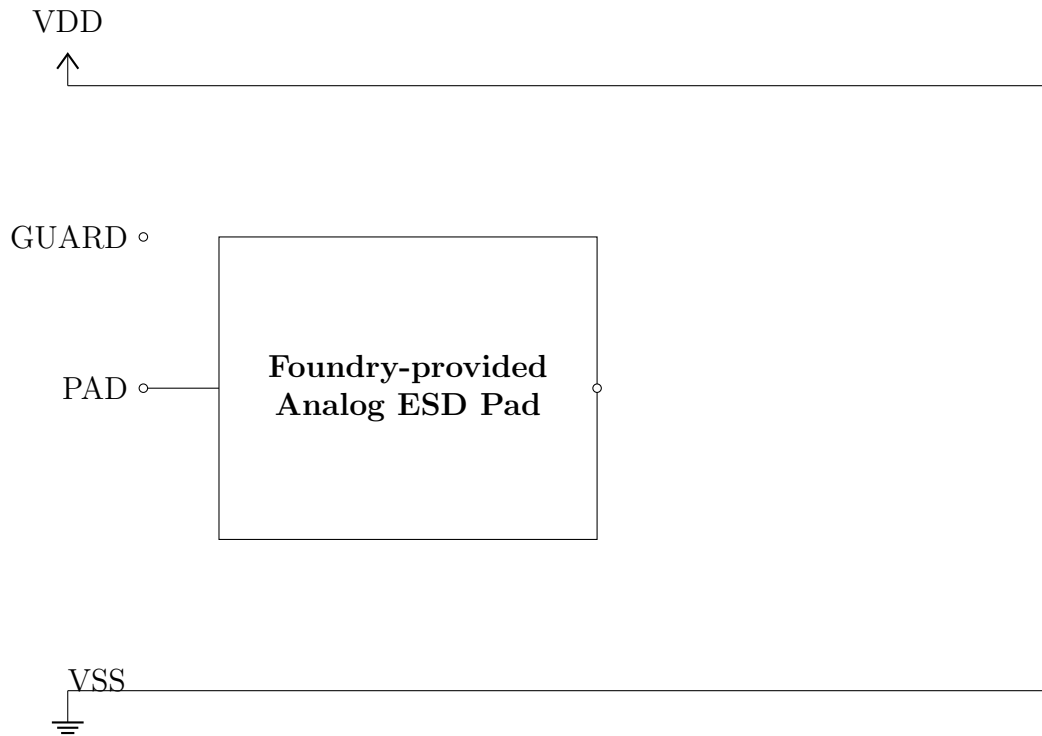
Implements the foundry provided analog pad with nothing connected to its output. The bondpad itself is guarded, but the internal structures are not, as they are left untouched. This allows for measuring the intrinsic leakage of the pad. It could also be used for ESD testing, to characterize the robustness of the pad structure itself.

7.12.2 Pins (array)

```
#71 Guard
#77 Hot input
#80 Guard
```

Guards are internally shorted together.

7.12.3 Diagram (abstract, horizontal)



7.13 CHARGE_PUMP

7.13.1 Purpose

Implements the fifty nifty pico ampere current source.

H. Pretl and M. Eberlein, IEEE SSC Magazine 2021 (DOI: 10.1109/MSSC.2021.3088968);

Its operation is described in detail in

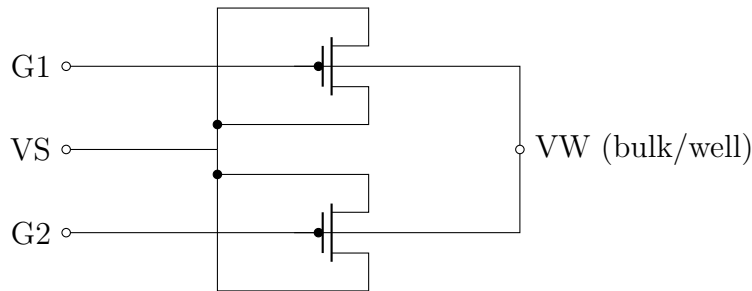
U. Cilingiroglu et al., IEEE JSSC 2003 (DOI: 10.1109/JSSC.2002.806282).

Two matched PMOS transistors with separate gates enabling opposite polarity operation to reduce injected noise. Implemented with PMOS W=40 μm , L=1 μm (provided).

7.13.2 Pins (array)

- #83 Guard
- #84 Gate of transistor 2
- #85 VS (drain and source of both transistors)
- #86 Gate of transistor 1
- #87 VW (bulk of both transistors)

7.13.3 Diagram



Guard ◦

8 Bonding Diagram

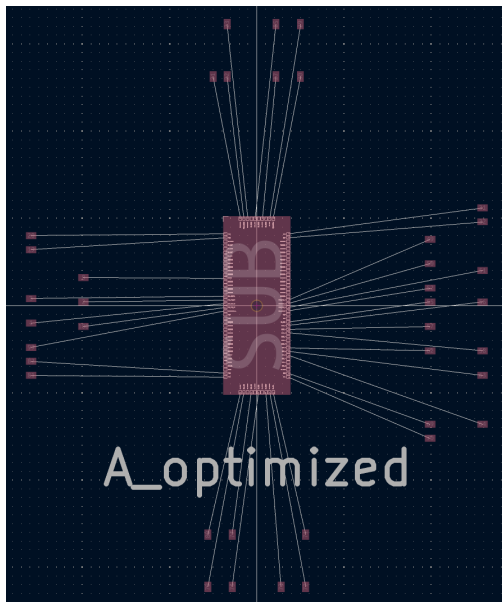
8.1 Die / Pad Dimensions

The ASIC measures $1936\text{ }\mu\text{m}$ by $5122\text{ }\mu\text{m}$. Its height is $250\text{ }\mu\text{m}$.

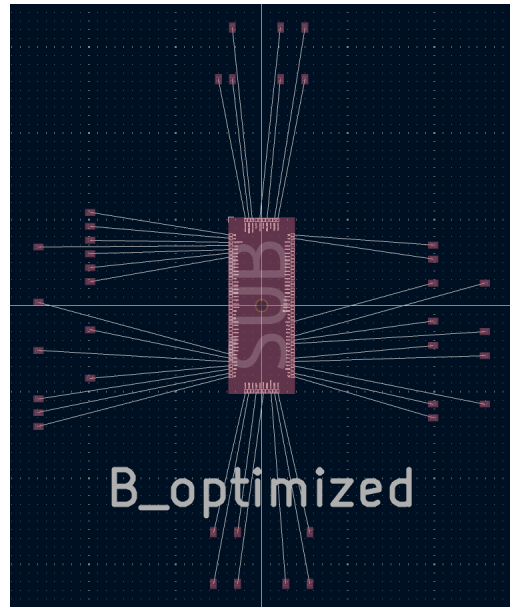
The bondpads measure $78\text{ }\mu\text{m}$ by $100\text{ }\mu\text{m}$. Their pitch is $105\text{ }\mu\text{m}$. The outward edge of the pad is $26\text{ }\mu\text{m}$ away from the sealring.

8.2 Bonding Diagram and Recommended PCB Footprint

As the ASIC is composed of many pads, and the ultra-low current nature of the teststructures require a certain spacing between the PCB pads, two bonding diagrams have been made, each one allowing access to a subset of the test structures.



(a) Bonding diagram variant A



(b) Bonding diagram variant B

Figure 7: trident-gf180-teststructures bonding diagrams

#	side	bondpad_x	bondpad_y	pcbpad_x	pcbpad_y	length_mm	angle [°]
1	WEST	-0.903	0.0525	-6.468	-0.5	5.592359	-5.67
2	WEST	-0.903	0.1575	-4.968	0.1	4.065407	-0.81
3	WEST	-0.903	0.2625	-6.468	0.2	5.565351	-0.64
8	WEST	-0.903	0.7875	-4.968	0.8	4.065019	0.18
19	WEST	-0.903	1.9425	-6.468	1.6	5.57553	-3.52
20	WEST	-0.903	2.0475	-6.468	2	5.565203	-0.49
21	NORTH	-0.4725	2.485	-1.25	6.561	4.149492	-10.8
22	NORTH	-0.3675	2.485	-0.85	6.561	4.104459	-6.75
23	NORTH	-0.2625	2.485	-0.85	8.061	5.606865	-6.01
25	NORTH	-0.0525	2.485	0.55	8.061	5.608456	6.17
27	NORTH	0.1575	2.485	0.55	6.561	4.094854	5.5
29	NORTH	0.3675	2.485	1.25	8.061	5.645404	8.99
30	NORTH	0.4725	2.485	1.25	6.561	4.149492	10.8
31	EAST	0.903	2.0475	6.468	2.8	5.615646	-7.7
32	EAST	0.903	1.9425	6.468	2.4	5.583774	-4.7
49	EAST	0.903	0.1575	4.968	1.9	4.422729	-23.2
50	EAST	0.903	0.0525	4.968	1.2	4.223859	-15.76
51	EAST	0.903	-0.0525	6.468	1	5.663654	-10.71
52	EAST	0.903	-0.1575	4.968	0.5	4.117831	-9.19
55	EAST	0.903	-0.4725	4.968	0.1	4.105116	-8.02
56	EAST	0.903	-0.5775	6.468	0.1	5.606089	-6.94
57	EAST	0.903	-0.6825	4.968	-0.6	4.065837	-1.16
58	EAST	0.903	-0.7875	6.468	-1.3	5.588549	5.26
62	EAST	0.903	-1.2075	4.968	-1.3	4.066052	1.3
63	EAST	0.903	-1.3125	6.468	-2	5.607306	7.04
64	EAST	0.903	-1.4175	6.468	-3.4	5.907583	19.61
69	EAST	0.903	-1.9425	4.968	-3.4	4.318395	19.73
70	EAST	0.903	-2.0475	4.968	-3.8	4.426678	23.32
71	SOUTH	0.4725	-2.485	1.4	-6.561	4.180195	-12.82
72	SOUTH	0.3675	-2.485	1.4	-8.061	5.670788	-10.49
73	SOUTH	0.2625	-2.485	0.7	-8.061	5.593137	-4.49
75	SOUTH	0.0525	-2.485	-0.7	-8.061	5.626547	7.69
77	SOUTH	-0.1575	-2.485	-0.7	-6.561	4.111944	7.58
79	SOUTH	-0.3675	-2.485	-1.4	-8.061	5.670788	10.49
80	SOUTH	-0.4725	-2.485	-1.4	-6.561	4.180195	12.82
81	WEST	-0.903	-2.0475	-6.468	-2	5.565203	0.49
82	WEST	-0.903	-1.9425	-6.468	-1.6	5.57553	3.52
99	WEST	-0.903	-0.1575	-6.468	-1.2	5.661805	-10.61
100	WEST	-0.903	-0.0525	-4.968	-0.6	4.101705	-7.67

Figure 8: Bonding wire information for variant A

#	side	bondpad_x	bondpad_y	pcbpad_x	pcbpad_y	length_mm	angle [°]
14	WEST	-0.903	1.4175	-4.968	0.7	4.127836	-10.01
15	WEST	-0.903	1.5225	-4.968	1.1	4.086898	-5.93
16	WEST	-0.903	1.6275	-4.968	1.5	4.066999	-1.8
17	WEST	-0.903	1.7325	-6.468	1.7	5.565095	-0.33
18	WEST	-0.903	1.8375	-4.968	1.9	4.06548	0.88
19	WEST	-0.903	1.9425	-4.968	2.3	4.08069	5.03
20	WEST	-0.903	2.0475	-4.968	2.7	4.117035	9.12
21	NORTH	-0.4725	2.485	-1.25	6.561	4.149492	-10.8
22	NORTH	-0.3675	2.485	-0.85	6.561	4.104459	-6.75
23	NORTH	-0.2625	2.485	-0.85	8.061	5.606865	-6.01
25	NORTH	-0.0525	2.485	0.55	8.061	5.608456	6.17
27	NORTH	0.1575	2.485	0.55	6.561	4.094854	5.5
29	NORTH	0.3675	2.485	1.25	8.061	5.645404	8.99
30	NORTH	0.4725	2.485	1.25	6.561	4.149492	10.8
31	EAST	0.903	2.0475	4.968	1.75	4.075872	4.19
32	EAST	0.903	1.9425	4.968	1.35	4.107953	8.29
55	EAST	0.903	-0.4725	4.968	0.65	4.217135	-15.44
59	EAST	0.903	-0.8925	6.468	0.65	5.774819	-15.49
60	EAST	0.903	-0.9975	4.968	-0.45	4.101705	-7.67
61	EAST	0.903	-1.1025	6.468	-0.75	5.576153	-3.62
65	EAST	0.903	-1.5225	4.968	-1.15	4.082032	-5.24
66	EAST	0.903	-1.6275	6.468	-1.45	5.56783	-1.83
67	EAST	0.903	-1.7325	6.468	-2.85	5.676093	11.35
69	EAST	0.903	-1.9425	4.968	-2.85	4.165067	12.58
70	EAST	0.903	-2.0475	4.968	-3.25	4.239131	16.48
71	SOUTH	0.4725	-2.485	1.4	-6.561	4.180195	-12.82
72	SOUTH	0.3675	-2.485	1.4	-8.061	5.670788	-10.49
73	SOUTH	0.2625	-2.485	0.7	-8.061	5.593137	-4.49
75	SOUTH	0.0525	-2.485	-0.7	-8.061	5.626547	7.69
77	SOUTH	-0.1575	-2.485	-0.7	-6.561	4.111944	7.58
79	SOUTH	-0.3675	-2.485	-1.4	-8.061	5.670788	10.49
80	SOUTH	-0.4725	-2.485	-1.4	-6.561	4.180195	12.82
81	WEST	-0.903	-2.0475	-6.468	-3.5	5.751433	-14.63
82	WEST	-0.903	-1.9425	-6.468	-3.1	5.684103	-11.75
83	WEST	-0.903	-1.8375	-6.468	-2.7	5.631441	-8.81
84	WEST	-0.903	-1.7325	-4.968	-2.1	4.081578	-5.17
85	WEST	-0.903	-1.6275	-6.468	-1.3	5.574628	3.37
86	WEST	-0.903	-1.5225	-4.968	-0.7	4.147376	11.44
87	WEST	-0.903	-1.4175	-6.468	0.1	5.768191	15.25

Figure 9: Bonding wire information for variant B